

Question 1

1. Steps

- a. `count = 0`
for each letter position:
 if curr letter != goal letter:
 count += 1
return count
- b. It lower bounds the number of actions left. With every legal action, only one letter can be changed at a time. Therefore, each position that differs from the goal must be changed at least once.
- c. It can't overestimate because there is a fixed number of positions that differ from the goal. To reach the goal state, each of those positions must be changed at least once. Since each step costs 1, the remaining cost must be at least the number of differing positions.

2. Heuristic

- a. `total = 0`
for each position i:
 if curr[i] != goal[i]:
 total += scrabble value of goal[i]
return total
- b. It lower bounds the cost of the scrabble inserting the goal letters that are not in the correct position. All wrong positions need to get their goal letter.
- c. It can't overestimate because the scrabble cost is based on the updated letter. So, each position that isn't the goal receives a cost of inserting to get to the goal letter. A path might make extra letter changes and cost even more, but it cannot avoid paying the cost of inserting those required goal letters.

3. Frequency

- a. `count = 0`
for each letter position:
 if curr letter != goal letter:
 count += 1
return count
- b. It lower bounds the unavoidable remaining frequency cost. For each position that isn't the goal needs at least one action with cost one because freq cost is one plus a nonnegative number stored in the dictionary for each word. So, the remaining cost is at least the number of positions from the goal.
- c. It can't overestimate because every action changes only one letter position and, under the frequency cost, every action costs at least 1 since values of nonnegative numbers stored in the dictionary are nonnegative. Therefore, if a fixed number of positions still differ from the goal, at least that many actions are required, so the true remaining frequency cost must be at least that amount.

Question 2

	A	B	C	D	E	F	G	
1	Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction	
2	WHY →ASK	Steps	5	773	5	16	97.93%	
3	WHY →ASK	Scrabble	9	765	9	13	98.30%	
4	WHY →ASK	Frequency	38.85540652	808	38.85540652	755	6.56%	
5	BEAR →DUCK	Steps	4	691	4	6	99.13%	
6	BEAR →DUCK	Scrabble	11	2981	11	6	99.80%	
7	BEAR →DUCK	Frequency	38.2402284	1084	38.2402284	720	33.58%	

1. UCS and A* returned the same optimal cost for all six.
2. Scrabble had the largest reduction for both WHY → ASK(98.30%) and BEAR → DUCK(99.80%).
3. For both, frequency was least informative WHY → ASK(6.56%) and BEAR → DUCK(33.58%).
4. Step heuristic counts the number of positions that differ from the goal, which estimates the least number of actions required. The scrabble heuristic is more informative because it adds the unavoidable cost of inserting the goal letters. The frequency heuristic only counts the differing positions even though each move actually costs $1 + \text{rarity}(w)$, where the rarity value is nonnegative and stored in the dictionary. Since the heuristic ignores the rarity portion of the cost, it can underestimate the remaining frequency cost by a large amount and therefore gives A* less guidance. This is why A* reduced less node expansions for frequency.

Question 3

1. Steps
 - a. Yes
 - b. Each legal move costs 1 and since the actions are reversible every path with W_{sub1} and W_{sub2} can be traced the opposite direction with the same number of steps and cost.
2. Scrabble
 - a. No
 - b. It is reversible but the cost is in correspondence to the letter inserted. going in the reverse direction can have a different cost. With words cat to sat inserting s costs 1, while sat to cat inserts c which costs 3. So the action is reversible, but the Scrabble cost is not the same.
3. Frequency
 - a. No
 - b. It is reversible but the cost is dependent on the word entered because a move from W_{sub1} to W_{sub2} costs $1 + \text{rarity}(W_{sub2})$ and the opposite $1 + R(W_{sub1})$. The rarity values are not always the same. With cat and sat, $1 + R(\text{sat})$ is 6.42999 and $1 + R(\text{cat})$ is 7.886906.

Question 4

1. **Gemini 3.1 Pro Prompt:** In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary. The baseline heuristic h_0

(n) is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

Response:

Proposed Heuristic Definition

Let $h_0(n)$ be the baseline number of positions where state n differs from the goal state g . Let $R(w)$ be the nonnegative rarity cost of entering word w , and let $N(n)$ be the set of valid dictionary words that can be reached from n by changing exactly one letter.

The proposed rarity-aware heuristic $h_1(n)$ is defined as:

$$h_1(n) = \begin{cases} 0 & \text{if } n = g \\ 1 + R(g)h_0(n) + R(g) + \min_{w \in N(n)} R(w) & \text{if } h_0(n) = 1 \\ h_0(n) & \text{if } h_0(n) \geq 2 \end{cases}$$

(Note: If a state n is a dead end and $N(n)$ is empty, the minimum over the empty set evaluates to ∞ , which correctly reflects an unsolvable state).

Justification for Admissibility ($h_1(n) \leq h^*(n)$)

To be admissible, the heuristic must never overestimate the true optimal cost $h^*(n)$. Because each transition changes exactly one letter and the cost to enter any word w is $1 + R(w) \geq 1$, any path of length m will cost at least m . A valid path from n to g requires a minimum of $m \geq h_0(n)$ transitions.

- **Case $n=g$:** The distance is 0, costing 0. $h_1(g)=0$.

- **Case $h_0(n)=1$:** The optimal path takes $m \geq 1$ transitions. The very final transition of any successful path must be into the goal state g , which strictly costs $1+R(g)$. Thus, the true cost $h^*(n) \geq 1+R(g) = h_1(n)$.
- **Case $h_0(n) \geq 2$:** The optimal path requires $m \geq h_0(n) \geq 2$ transitions.
 1. The **first transition** out of state n must enter some neighbor word $w_1 \in N(n)$. The cost of this single step is $1+R(w_1)$, which is bounded below by $1+\min_{w \in N(n)} R(w)$.
 2. The **final transition** must enter the goal state g , costing exactly $1+R(g)$.
 3. Because $h_0(n) \geq 2$, g is at least two letter changes away, meaning $g \notin N(n)$. This logically guarantees that the first transition and the final transition are **distinct steps** in the sequence.
 4. The remaining $m-2$ intermediate transitions each cost at least 1 (since $R(w) \geq 0$).
- Summing the minimum costs of these components yields:
 Total Cost $\geq [1+\min_{w \in N(n)} R(w)] + [1+R(g)] + (m-2) \times 1$
 Total Cost $\geq m + R(g) + \min_{w \in N(n)} R(w)$
 Because $m \geq h_0(n)$, we substitute $h_0(n)$ for m to establish the final admissible lower bound, which exactly matches $h_1(n)$.

Justification for Being More Informative

A heuristic is strictly more informative if it dominates the baseline without overestimating the true cost.

Because dictionary rarities are nonnegative, both $R(g)$ and $\min_{w \in N(n)} R(w)$ are ≥ 0 . Therefore, $h_1(n) \geq h_0(n)$ is mathematically guaranteed for every possible state.

Furthermore, $h_1(n)$ will be **strictly larger** than $h_0(n)$ for at least one state in any non-trivial scenario. As long as the targeted goal word g has a rarity $R(g) > 0$, the heuristic guarantees $h_1(n) > h_0(n)$ across the entire search space. Even in the worst-case scenario where $R(g) = 0$, $h_1(n)$ remains strictly larger for any state n where all immediately accessible neighbors possess a rarity greater than 0.

2.

$$h_1(n) = \begin{cases} 0 & \text{if } n = g \\ 1 + R(g) & \text{if } h_0(n) = 1 \\ h_0(n) + R(g) + \min_{w \in N(n)} R(w) & \text{if } h_0(n) \geq 2 \end{cases}$$

3.

The heuristic is admissible because it has remaining cost 0 and is in goal state. Arriving at the goal state needs costs $R(g)$ plus 1 or $h_{sub0}(n) = 1$. If i is equal or greater to 2 all solutions need $h_{sub0}(n)$ actions. The first move needs some legal neighbor that will cost at least 1 plus $\min_{w \in N(n)} R(w)$, and the final must cost $R(g)$ plus 1. The start and last action will be different when there are at least two positions that are different and the rest of the actions cost at least 1. So, the heuristic is admissible because it does not overestimate the remaining cost.

4. The proposal is not invalid.

5. Heuristic h_{sub1} returned the same optimal solution cost as h_{sub0} for both frequency cost problems, but expanded fewer nodes in each. As for why to ask, the node expanded from 755 to 497. As for bear to duck, 720 to 201. This makes h_{sub1} better because it was most informative and had the most optimal solution cost.